Configuring JS8Call to Use rigctld with FLRig on Linux Mint

A step-by-step guide for Linux users

This guide shows how to install Hamlib's rigctld service and configure it as an intermediary between JS8Call and FLRig. The goal is to provide JS8Call with a dedicated network CAT connection to FLRig, which can help isolate or reduce CAT/PTT delays seen with some direct JS8Call-to-FLRig configurations.

What you will build: JS8Call will connect to rigctld on this computer. rigctld will then connect to FLRig, and FLRig will continue controlling the radio.	
Control path	JS8Call -> rigctld -> FLRig -> radio
JS8Call rig type	Hamlib NET rigctl
JS8Call server address	localhost:4539
rigctld FLRig backend	Hamlib model 4
FLRig server address	127.0.0.1:12345 (unless you changed FLRig's port)

Before You Begin

This guide assumes the following are already true:

- Linux Mint is installed and you can log in to the desktop.
- FLRig is installed and can already control your radio.
- JS8Call is installed.
- Your radio's audio connection to JS8Call is already working, or will be configured separately.

You do not need prior Linux command-line experience. Every terminal command required by this procedure is shown exactly as it should be entered.

Important: CAT Control and Audio Are Separate

rigctld handles radio control commands such as frequency, mode, and PTT. It does not carry JS8Call audio. Continue using your normal Linux audio setup, such as PulseAudio/PipeWire-Pulse and PulseAudio Volume Control (pavucontrol), for receive and transmit audio.

Step 1 - Open a Terminal

A terminal is a text window used to enter Linux commands. In Linux Mint, press Ctrl+Alt+T. A terminal window should open with a prompt similar to:

```
yourname@yourcomputer:~$
```

Do not type the example prompt. Type only the commands shown in the command blocks below.

Step 2 - Install rigctld

rigctld is included in the Hamlib utilities package. You do not need to download rigctld manually from a website.

In the terminal, enter:

```
sudo apt update
```

Press Enter. Linux may ask for your account password. When typing a password in a terminal, no letters, dots, or asterisks appear on the screen. This is normal. Type the password and press Enter.

When the update finishes, enter:

```
sudo apt install libhamlib-utils
```

If Linux asks whether you want to continue, type Y and press Enter.

What this installs: The package includes rigctld, which will act as the server between JS8Call and FLRig, and rigctl, a command-line utility used later to test the connection.

Step 3 - Verify the Installation

Enter these two commands, one at a time:

```
rigctld --version
rigctl --version
```

Each command should display Hamlib version information. If Linux instead says 'command not found', repeat Step 2 and confirm the installation completed successfully.

Step 4 - Start FLRig and Confirm It Works

1. Start FLRig the same way you normally do.
2. Confirm FLRig displays the correct radio frequency.
3. Change frequency slightly in FLRig and verify that the radio follows.
4. If you normally use FLRig for PTT, briefly verify that PTT also works.

Do not continue until FLRig can communicate normally with the radio. rigctld depends on a working FLRig connection.

Step 5 - Confirm FLRig's Network Server

rigctld communicates with FLRig through FLRig's network/XML-RPC server. The commonly used FLRig server port is 12345.

In FLRig, open the configuration/preferences area and confirm that its XML-RPC/network server is enabled. If the port is set to something other than 12345, write that number down. You will substitute your port wherever this guide shows 12345.

Beginner checkpoint: At this point FLRig should be running and connected to the radio. Leave FLRig running for the remaining steps.

Step 6 - Test Hamlib Directly Against FLRig

Before creating the permanent launcher, test whether Hamlib can reach FLRig.

With FLRig still running, enter:

```
rigctl -m 4 -r 127.0.0.1:12345 f
```

If your FLRig server uses a different port, replace 12345 with that port.

A successful result normally prints the radio frequency in Hz. For example, 14115000 represents 14.115 MHz.

If this test fails: Do not continue to JS8Call yet. Recheck that FLRig is running, that its network/XML-RPC server is enabled, and that the port number is correct.

Step 7 - Create the rigctld Launcher Script

The following commands create a small launcher file on your Desktop. The launcher starts rigctld with the correct settings for JS8Call.

Copy and paste the entire block into the terminal:

```
cat > ~/Desktop/Rigctld-JS8.sh <<'EOF'
#!/usr/bin/env bash
exec rigctld -m 4 -r 127.0.0.1:12345 -T 127.0.0.1 -t 4539 -vvv
EOF

chmod +x ~/Desktop/Rigctld-JS8.sh
```

If your FLRig server uses a port other than 12345, open the file later with a text editor and replace 12345 with the correct port.

What the rigctld Command Means

Part	Meaning
rigctld	Starts the Hamlib radio-control server.
-m 4	Uses Hamlib model 4, the FLRig backend.
-r 127.0.0.1:12345	Connects rigctld to FLRig on this same computer.
-T 127.0.0.1	Allows connections only from this computer instead of exposing rigctld to the local network.
-t 4539	Makes rigctld listen for JS8Call on TCP port 4539.
-vvv	Shows diagnostic information in the terminal, useful for troubleshooting.

Why Port 4539?

Port 4539 is used in this guide so it does not conflict with other rigctld instances that may already be using ports such as 4532 or 4537. Any unused TCP port can be used, but the same port must be entered in both the rigctld launcher and JS8Call.

Hamlib convention normally associates rigctld with even-numbered ports and rotctld with odd-numbered ports. Port 4539 still works technically. If you prefer to follow the convention, use an unused even port such as 4538 or 4540 and substitute it everywhere in this guide.

Step 8 - Start rigctld

The most reliable beginner method is to start the script from a terminal:

```
~/Desktop/Rigctld-JS8.sh
```

The terminal should remain open and begin showing Hamlib diagnostic messages. Leave this terminal running while using JS8Call.

Do not close this terminal: Closing the rigctld terminal stops the bridge between JS8Call and FLRig.

Step 9 - Verify rigctld Is Listening

Open a second terminal by pressing Ctrl+Alt+T again, then enter:

```
ss -ltnp | grep 4539
```

A successful result should show 127.0.0.1:4539 in a LISTEN state. This means rigctld is waiting for a local program such as JS8Call to connect.

Step 10 - Test the Complete Bridge

In the second terminal, enter:

```
rigctl -m 2 -r 127.0.0.1:4539 f
```

A successful response should print the radio frequency in Hz.

This single test confirms the complete control chain:

```
rigctl test client -> rigctld -> FLRig -> radio
```

Checkpoint: Do not configure JS8Call until this test succeeds. If this test works, the rigctld-to-FLRig portion of the setup is functioning.

Step 11 - Create a Separate JS8Call Configuration

Using a separate JS8Call configuration lets you test this setup without overwriting your existing working settings.

1. Start JS8Call.
2. Open the Configurations menu.
3. Clone your normal or default working configuration.
4. Give the clone a descriptive name, such as FLRig via rigctld.
5. Switch to the new cloned configuration.

JS8Call may restart after switching configurations. That is normal.

Step 12 - Configure JS8Call CAT Control

In JS8Call, open:

```
File -> Settings -> Radio/CAT Control
```

Set the following values:

JS8Call setting	Value
Rig	Hamlib NET rigctl
Network Server	localhost:4539

The exact location or wording of these fields can vary slightly by JS8Call version, but the key requirement is that the selected rig is Hamlib NET rigctl and the server is localhost:4539.

Important: Do not select JS8Call's direct FLRig rig option in this configuration. JS8Call must connect to the external rigctld server instead.

Step 13 - Test CAT

1. Click Test CAT in JS8Call.
2. The test should report success.
3. Confirm that JS8Call displays the correct radio frequency.
4. If possible, change frequency and confirm that JS8Call, FLRig, and the radio remain synchronized.

Step 14 - Test PTT

1. Click or toggle Test PTT ON.
2. Confirm that the radio keys promptly.
3. Toggle Test PTT OFF.
4. Confirm that the radio unkeys promptly.
5. Click OK to save the JS8Call settings.

Keep the rigctld terminal visible during this test. The diagnostic output can help identify where a delay occurs.

Step 15 - Verify JS8Call Audio

Because rigctld changes only CAT/PTT control, verify your normal JS8Call audio routing separately.

1. Open PulseAudio Volume Control (pavucontrol) if you normally use it.
2. Confirm JS8Call receive audio is coming from the correct radio or USB audio input.
3. Confirm JS8Call transmit audio is routed to the correct radio or USB audio output.
4. Confirm JS8Call is not accidentally using a Monitor source unless a Monitor source is intentionally required by your setup.

Normal Startup Procedure

After initial setup, start the applications in this order:

1. Start your normal audio tools, such as pavucontrol, if needed.
2. Start FLRig and wait until it is communicating with the radio.
3. Start rigctld by running ~/Desktop/Rigctld-JS8.sh.
4. Leave the rigctld terminal open.
5. Start JS8Call using the FLRig via rigctld configuration.
6. Verify frequency and CAT control before operating.

Normal Shutdown Procedure

1. Exit JS8Call.
2. Return to the rigctld terminal and press Ctrl+C.
3. Exit FLRig.
4. Close other audio utilities if desired.

Optional - Launch rigctld by Double-Clicking an Icon

Once the terminal-based procedure is working, you can create a desktop launcher. This is optional; test the basic setup first.

In a terminal, first find your Linux username:

```
whoami
```

Then create the launcher:

```
cat > ~/Desktop/Rigctld-JS8.desktop <<'EOF'
[Desktop Entry]
Type=Application
Name=Rigctld - JS8Call
Comment=Start the JS8Call to FLRig rigctld bridge
Exec=/home/YOUR_USERNAME/Desktop/Rigctld-JS8.sh
Terminal=true
Icon=utilities-terminal
Categories=HamRadio;
EOF

chmod +x ~/Desktop/Rigctld-JS8.desktop
```

Replace YOUR_USERNAME with the username shown by the whoami command. On some Linux Mint systems you may need to right-click the icon and choose Allow Launching.

Troubleshooting

Problem: rigctld says 'command not found'

Install the Hamlib utilities again:

```
sudo apt update
sudo apt install libhamlib-utils
```

Problem: Direct FLRig test fails

Run:

```
rigctl -m 4 -r 127.0.0.1:12345 f
```

- Make sure FLRig is running.
- Make sure FLRig is already controlling the radio.
- Make sure the FLRig XML-RPC/network server is enabled.
- Confirm the FLRig server port. If it is not 12345, use the actual port.

Problem: Nothing is listening on port 4539

Check:

```
ss -ltnp | grep 4539
```

- If there is no output, rigctld is not running.
- Return to the rigctld terminal and look for an error message.
- Restart FLRig first, then restart rigctld.

Problem: Port 4539 is already in use

Find the process using it:

```
ss -ltnp | grep 4539
```

Choose another unused port, such as 4538 or 4540. Change the -t value in the launcher script and change JS8Call's Network Server to the same port.

Problem: JS8Call Test CAT fails

- Confirm rigctld is still running.
- Confirm JS8Call Rig is Hamlib NET rigctl.
- Confirm Network Server is localhost:4539.
- Repeat the terminal test: rigctl -m 2 -r 127.0.0.1:4539 f
- If the terminal test succeeds but JS8Call fails, the problem is likely in the JS8Call CAT configuration.

Problem: CAT works but PTT is slow

Watch the rigctld terminal while toggling Test PTT. The diagnostic output helps separate three possible locations for the delay:

- JS8Call delay: rigctld does not receive the PTT request promptly.
- rigctld/FLRig delay: rigctld receives the request promptly but waits for FLRig.
- Radio-side delay: FLRig processes the request but the radio responds slowly.

If the issue is being investigated because of Linux CPU starvation or late audio callbacks, remember that audio scheduling and CAT control are separate subsystems. A busy CPU can still delay either application even though rigctld itself is configured correctly.

Problem: Audio works poorly after this change

This procedure does not intentionally change audio. Recheck pavucontrol/PipeWire-Pulse device selections and JS8Call audio input/output settings. Do not change CAT settings to troubleshoot an audio routing problem.

How to Return to the Previous JS8Call Setup

1. Exit JS8Call.
2. Stop rigctld by pressing Ctrl+C in its terminal.
3. Restart JS8Call.
4. Switch back to your original JS8Call configuration.

Because the original JS8Call configuration was cloned instead of modified, the previous settings remain available.

Quick Reference

Item	Setting
Install package	sudo apt install libhamlib-utils
FLRig server	127.0.0.1:12345
rigctld backend	Hamlib model 4
rigctld listen address	127.0.0.1
rigctld port	4539
JS8Call rig	Hamlib NET rigctl
JS8Call Network Server	localhost:4539
Test FLRig directly	rigctl -m 4 -r 127.0.0.1:12345 f
Test complete bridge	rigctl -m 2 -r 127.0.0.1:4539 f

Technical References

Hamlib rigctl documentation: https://hamlib.sourceforge.net/snapshots/man1/rigctl.1.html

Hamlib rigctld documentation: https://hamlib.sourceforge.net/snapshots/man1/rigctld.1.html

Ubuntu package information for libhamlib-utils: https://packages.ubuntu.com/libhamlib-utils